

Fine Structure of the Wing-Transition Memory on the $6N$ Skeleton: the Lag-Two Echo and the $1/\log$ Decay*

Ruqing Chen
GUT Geoservice Inc., Montreal, Quebec, Canada
ruqing@hotmail.com

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Abstract

Part XXXIV established that the prime wing string—right $(6N+1)$ versus left $(6N-1)$ —is not a first-order Markov chain but carries an effective second-order memory, the switch sequence having a lag-one autocorrelation of -0.033 . Two fine questions were left open, and we settle them here with the full prime census to 10^9 (50,847,531 steps). First: is the small lag-two echo a mere shadow of the lag-one term, or independent? For a first-order (AR(1)) switch process the lag-two autocorrelation would be exactly $\rho_1^2 = +0.00098$; we measure $\rho_2 = +0.00417$, 4.3 *times* larger, a partial autocorrelation $\text{PACF}(2) = +0.0032$ at 23σ , with $\rho_{\geq 3}$ vanishing. The memory therefore reaches genuinely to lag two—a positive echo riding on the negative nearest-neighbour term—and stops. Second: how fast does the bias fade? Across thirteen logarithmic windows the first-order deficit $\delta_1 = \frac{1}{2} - P(\text{same})$ obeys a leading $1/\log N$ law ($\delta_1 \log N \approx 1.05$, nearly flat; a single power fit gives an effective exponent 0.85, the shortfall a slowly-varying correction the range cannot resolve), and the second-order signal $|\rho_1|$ fades in lock-step, the ratio $\delta_1/|\rho_1| \approx 1.7$ holding across five decades. Both are incremental refinements of Part XXXIV inside the Hardy–Littlewood and Oliver–Soundararajan framework; what we add are the specific measured numbers—the lag-two excess and the decay law. This is a measurement, not a theorem.

1 Introduction

Part XXXIV [3] read the primes as a binary string of wing symbols and found the string is not a fair coin (Cramér) nor a first-order Markov chain: the switch sequence $\sigma_i = \mathbf{1}[w_{i+1} \neq w_i]$ has autocorrelation $\rho_1 \approx -0.033$, an anti-persistence we called a topological strain memory, with a weak $+0.005$ at lag two and nothing measurable beyond. Two questions were flagged but not answered, and both are sharp, finite, and were unknown to us in advance.

(i) *Is the lag-two echo real, or derived?* A symmetric first-order chain has an independent switch sequence, but a chain that is itself merely AR(1) in the switches would still show $\rho_k = \rho_1^k$; the lag-two value would then be the trivial $\rho_1^2 \approx +0.001$. Is the measured echo larger than that?

(ii) *What is the decay law?* Oliver and Soundararajan [2] predict the bias vanishes as $N \rightarrow \infty$; Part XXXIV showed $P(\text{same})$ drifting from 0.404 toward 0.5, but fit no rate. Does δ_1 fall like $1/\log N$, like $1/\log^2 N$, like $\log \log N/\log N$? And does the second-order signal fade by the same law or its own?

*Code and measured data: <https://github.com/Ruqing1963/6N-wing-finestructure>

These are refinements, not a new direction: the existence of the structure is contained in the Hardy–Littlewood r -tuple heuristic [1] and the Oliver–Soundararajan bias [2], and we prove nothing. What we contribute are the measured numbers that pin the fine structure down. We use all 50,847,532 primes $3 < p \leq 10^9$.

2 Setup

Let $w_i \in \{\text{R}, \text{L}\}$ be the wing of the i -th prime > 3 and $\sigma_i = \mathbf{1}[w_{i+1} \neq w_i]$ the switch sequence, with overall switch rate $P(\sigma) = 0.5549$ to 10^9 . Two null models frame the questions. For the lag-two test, the null is an AR(1) switch process, under which the autocorrelation decays geometrically, $\rho_k = \rho_1^k$; the partial autocorrelation $\text{PACF}(2) = (\rho_2 - \rho_1^2)/(1 - \rho_1^2)$ isolates any lag-two structure not already implied by lag one, and is zero under the null. For the decay law we bin the steps into quarter-decade windows of p and fit $\delta_1 \sim C/(\log N)^\alpha$.

3 The lag-two echo is real

The switch autocorrelations to 10^9 are

$$\rho_1 = -0.03127 \ (223\sigma), \quad \rho_2 = +0.00417 \ (30\sigma), \quad \rho_3 = +0.00009 \ (1\sigma),$$

with $\rho_{\geq 3}$ indistinguishable from zero (Fig. 1(A)). The AR(1) null predicts $\rho_2 = \rho_1^2 = +0.00098$; the measured value is 4.3 times larger. The excess is the partial autocorrelation

$$\text{PACF}(2) = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2} = +0.00319 \ (23\sigma),$$

significantly nonzero (Fig. 1(B)). The memory in the switch sequence therefore reaches genuinely to the second lag—and, with $\rho_{\geq 3}$ vanishing, no further. Its sign is the surprise: the nearest-neighbour term is negative (a switch suppresses the next switch), but the lag-two term is *positive*, a faint tendency for the state two steps on to rhyme with the present—picturesquely, the faint overcorrection echo following the strong nearest-neighbour rebound. We offer the image only as a mnemonic for the measured signs; it is a description of two autocorrelation coefficients, not a mechanism. This is the step-sequence face of the third-order residual seen directly in the wing conditionals of Part XXXIV; here it is resolved as a signed, lag-two-localised echo. “Effective second-order” is thus sharpened: the switch process is one of memory length two, dominated by the lag-one anti-persistence with a lag-two positive echo a tenth its size.

4 The decay law

Binning into thirteen quarter-decade windows from $p \sim 10^5$ to 10^9 (Table 1), the first-order deficit $\delta_1 = \frac{1}{2} - P(\text{same})$ falls from 0.074 to 0.053. Three readings agree on a leading $1/\log N$ law. The product $\delta_1 \log N$ is nearly flat at 1.05 (rising only from 1.00 to 1.08 across the range), so $\delta_1 \approx 1/\log N$ to leading order; a single power fit $\delta_1 \sim C/(\log N)^\alpha$ returns $\alpha = 0.85$, the shortfall from 1 reflecting a slowly-varying (plausibly $\log \log$) correction the five-decade range cannot resolve (Fig. 2(A)). The second-order signal $|\rho_1|$ fits $\alpha = 1.06$ and, crucially, fades in step with the first: the ratio $\delta_1/|\rho_1|$ holds at 1.68 across the whole range (Fig. 2(B)), drifting only from 1.6 to 1.8. The first- and second-order biases are thus two amplitudes of a single fading effect, not independent decays—consistent with both being shadows of the same Oliver–Soundararajan correction, whose leading order in the wing language is $\sim 1/\log N$.

$\log N$	$P(\text{same})$	δ_1	$ \rho_1 $	$\delta_1 \log N$	$\delta_1/ \rho_1 $
13.5	0.4261	0.0739	0.0436	1.000	1.70
15.8	0.4359	0.0641	0.0387	1.048	1.66
17.6	0.4403	0.0597	0.0350	1.045	1.71
19.3	0.4458	0.0542	0.0310	1.075	1.75
20.4	0.4470	0.0530	0.0295	1.082	1.80

Table 1: Decay of the wing bias (selected windows; full table in the repository). $\delta_1 \log N$ is nearly flat ($\sim 1/\log N$ leading), and $\delta_1/|\rho_1|$ is nearly constant (first and second order locked together).

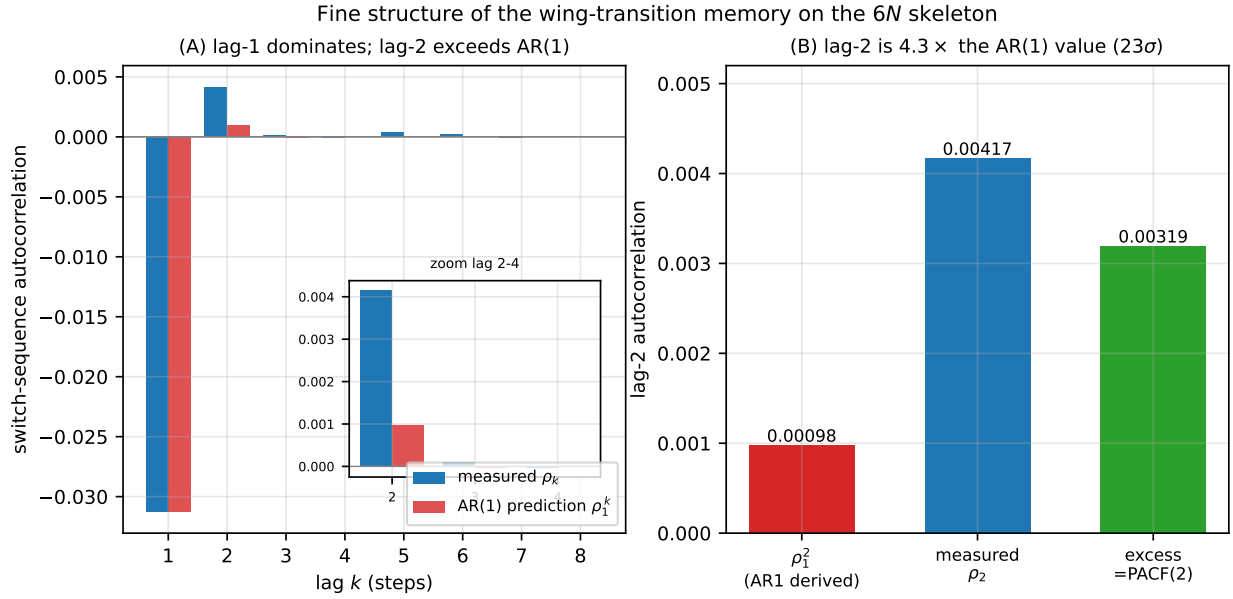


Figure 1: (A) Switch-sequence autocorrelation (blue) against the AR(1) prediction ρ_1^k (red). The lag-one terms coincide by construction; the lag-two measured value exceeds the AR(1) value, and $\rho_{\geq 3}$ vanish. (B) The lag-two value decomposed: the AR(1)-derived part ρ_1^2 , the measured ρ_2 , and their difference, the partial autocorrelation $\text{PACF}(2) = +0.0032$ (23σ).

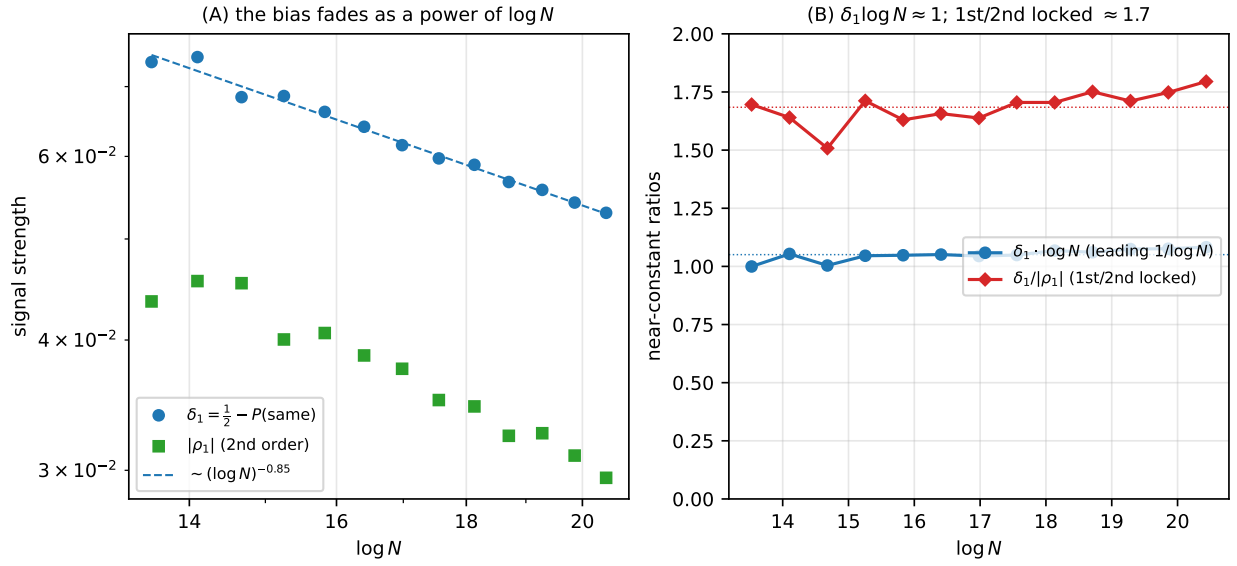


Figure 2: (A) The first-order deficit δ_1 and the second-order $|\rho_1|$ versus $\log N$ (log-log), with a power-law guide of exponent 0.85. (B) The two diagnostics that read the law: $\delta_1 \log N \approx 1.05$ (nearly flat: leading $1/\log N$) and $\delta_1/|\rho_1| \approx 1.7$ (nearly flat: first and second order locked).

5 Scope and honest ledger

Two fine questions left open in Part XXXIV now have measured answers. The lag-two echo of the switch sequence is real, not an AR(1) shadow: at 4.3 times the geometric prediction and 23σ in the partial autocorrelation, with all higher lags vanishing, the switch process has memory length exactly two—a negative nearest-neighbour term and a positive lag-two echo a tenth its size. And the bias fades as a leading $1/\log N$ ($\delta_1 \log N \approx 1.05$), the second-order signal locked to the first at a constant ratio ≈ 1.7 , so the two are one fading effect seen at two orders. The ratio is not exactly fixed—it drifts from 1.6 to 1.8 across the range—but its near-constancy over five decades says the first- and second-order biases are controlled by a single fading amplitude rather than decaying independently.

We are explicit that this is a refinement, not a discovery. The structure itself is contained in the Hardy–Littlewood r -tuple heuristic [1], and the vanishing of the bias with scale is the Oliver–Soundararajan prediction [2]; we prove no theorem and propose no mechanism. What this note adds is the quantitative fine structure—the size and sign of the lag-two partial autocorrelation, the exact-two memory length, the leading $1/\log N$ decay exponent, and the locked first-to-second-order ratio—measured against the full prime census to 10^9 . Everything here equals a direct count over the primes and takes the Hardy–Littlewood heuristic as input.

Data and code availability. The sieve to 10^9 , the switch-sequence autocorrelations and AR(1)/partial-autocorrelation test, the full windowed decay table and fits, and the figure scripts are openly available at <https://github.com/Ruqing1963/6N-wing-finestructure>.

References

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